Multimedia Signal processing

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1. **Question (8 pts)**:
   Consider two discrete memoryless sources X and Y, whose joint p.d.f. is given by

	b_1	b_2	$p(a)$
a_1	0.8	0.02	0.82
a_2	0.03	0.15	0.18
$p(b)$	0.83	0.17	1.0

   (a) Determine whether X and Y are statistically dependent sources.

   (b) Compute a lower bound on the average length of a code used to encode: i) X and Y independently; ii) X and Y jointly.

   (c) Consider the following sequences of symbols

source	symbols
X	a_1, a_1, a_2, a_1
Y	b_1, b_1, b_2, b_2

   An arithmetic encoder is designed to encode the symbols emitted by the source X, followed by those of the source Y. Compute the number of bits used by such arithmetic coder in two cases: i) neglecting, or ii) exploiting the statistical dependency between the sources.

   Solution

   (a) Two sources are statistically independent if and only if the joint p.m.f. can be factored out as the product of the marginal p.m.f.'s. In this case,

$$p(a_1, b_1) \neq p(a_1) \cdot p(b_1) \tag{1}$$

   Hence, X and Y are statistically dependent.

   (b) The average length of a code used to encode X is lower bounded by the entropy $H(X)$. That is,

$$L_X \geq H(X) = -0.82 \log_2 0.82 - 0.18 \log_2 0.18 = 0.62 \text{ bits / symbol} \tag{2}$$

   Similarly, for the source Y,

$$L_Y \geq H(Y) = -0.83 \log_2 0.83 - 0.17 \log_2 0.17 = 0.66 \text{ bits / symbol} \tag{3}$$

   By jointly encoding the sources, the average length is lower bounded by

$$L_{XY} \geq \frac{H(X,Y)}{2} = \frac{1}{2}(-0.8\log_2 0.8 - 0.02\log_2 0.02 - 0.15\log_2 0.15 - 0.03\log_2 0.03) = \frac{0.93}{2} = 0.47 \text{ bits / symbol} \tag{4}$$

(c) Encoding the symbols emitted by X requires to use a number of bits equal to

$$R_X = \lceil -\log_2(p(a_1)\cdot p(a_1)\cdot p(a_2)\cdot p(a_1))\rceil + 1 = 5 \tag{5}$$

Encoding the symbols emitted by X requires to use a number of bits equal to

$$R_Y = \lceil -\log_2(p(b_1)\cdot p(b_1)\cdot p(b_2)\cdot p(b_2))\rceil + 1 = 7 \tag{6}$$

Encoding the symbols emitted by Y can take advantage of the availability of the symbols emitted by X both at the encoder and at the decoder.

$$R_{Y|X} = \lceil -\log_2(p(b_1|a_1)\cdot p(b_1|a_1)\cdot p(b_2|a_2)\cdot p(b_2|a_1))\rceil + 1 \tag{7}$$

To compute $R_{Y|X}$, we need to determine the conditional probabilities

$$p(b_1|a_1) = p(a_1,b_1)/p(a_1) = 0.8/0.82 = 0.9756 \tag{8}$$
$$p(b_2|a_1) = p(a_1,b_2)/p(a_1) = 0.02/0.82 = 0.0244 \tag{9}$$
$$p(b_1|a_2) = p(a_2,b_1)/p(a_2) = 0.03/0.18 = 0.1667 \tag{10}$$
$$p(b_2|a_2) = p(a_2,b_2)/p(a_2) = 0.15/0.18 = 0.8333 \tag{11}$$

Hence,

$$R_{Y|X} = \lceil -\log_2(0.9756\cdot 0.9756\cdot 0.8333\cdot 0.0244)\rceil + 1 = 7 \tag{12}$$

2. **Question (5 pts)**

Discuss the design of the filter bank in audio coding.

Solution

See slide set `MMSP_2.2_Audio coding`.

3. **Question (5 pts)**

(a) Illustrate the properties of speech signals, both in the temporal and in the frequency domain.

(b) Describe the source-system model used to synthesize speech signals.

Solution

See slide set `MMSP_2.1_Speech_Coding`.

4. **Questions (5 pts - each answer can be either TRUE or FALSE)**

   (a) Consider a transform coding system that processes blocks of N input samples.

   - [T] [F] When the KLT transform is used, the transform coefficients are statistically independent.
     FALSE. The transform coefficients are uncorrellated. In general, this does not imply statistical independence.
   - [T] [F] Let $G_{TC}(N)$ denote the transform coding gain achievable using blocks of N samples. Then, $\lim_{N\to\infty} G_{TC}(N) = +\infty$.
     FALSE. $\lim_{N\to\infty} G_{TC}(N) = G_{TC}(\infty) < +\infty$. The value of $G_{TC}(\infty)$ depends on the nature of the memory of the source.
   - [T] [F] $R_{TC}(D) \leq R_{PCM}(D)$, where $R_{TC}(D)$ and $R_{PCM}(D)$ are, respectively, the operational rate-distortion curves of transform coding and scalar quantization.
     TRUE.
   - [T] [F] The asymptotic time complexity of computing the output of a transform coder is $O(N)$.
     FALSE. Computing the transform coefficients requires $O(N^2)$. The cost of quantization grows linearly with the number of samples ($O(N)$). Hence, the overall complexity is $O(N^2)$.

   (b) Consider a quantizer based on compander-expander

   - [T] [F] $D_{CE}(R) \leq D_U(R)$, where CE and U refer to, respectively, the compander-expander and uniform quantizer.
     TRUE.
   - [T] [F] In a compander-expander quantizer, the reconstruction levels are equally likely.
     TRUE. The p.d.f. of $y = T(x)$ is uniform. Applying a uniform scalar quantizer to y, the reconstruction levels are equally likely.
   - [T] [F] A compander-expander quantizer can be conveniently used to encode speech signals.
     TRUE. The p.d.f. of speech signals can be approximated as a Laplacian distribution. The c.d.f. of the Laplacian has been approximated in the past using the A-law and μ-law.

   (c) Consider an image coding system.

   - [T] [F] Setting the same quantization step size for all transform coefficients leads to a fixed number of bits for all blocks.
     FALSE. The number of bits of each block depends on the quantized transform coefficients and, in general, it varies for each block.
   - [T] [F] Let R denote the rate to encode an image. If $R_2 > R_1$, then $PSNR(R_1) > PSNR(R_2)$.
     FALSE. The rate-distortion curve PSNR vs. rate is monotone increasing.
   - [T] [F] Let R_I denote the rate to encode image I. If $R_I = R_J$, then $PSNR_I(R_I) = PSNR_J(R_J)$.
     FALSE. At the same target rate, two images I and J might be compressed at different distortion.

5. **Question (Matlab - 10pts)**

   Consider the image *image.png*.

   (a) Read the image pixels column-wise into a single vector.

   (b) PCM encode the pixel values.

   (c) DPCM encode the pixel values.

(d) Plot the rate-distortion function $PSNR(R)$ in both cases by varying the quantization step size. Use the entropy of the quantization symbols as an estimate of the rate.

Solution

```
%% 2014_02_07_MMSP_exam

clear all
close all
clc

%% a) Consider the image 'image.png'. Read the pixels column-wise into a single vector.
X = imread('image.png');
[nR, nC] = size(X);
X = double(X(:));


%% b) PCM encode the pixel values
Q = 200:-10:10;
R_pcm = zeros(length(Q),1);
PSNR_pcm = zeros(length(Q),1);
for ii = 1:length(Q)
    % quantize the signal
    X_tilde_pcm = Q(ii) * floor(X/Q(ii)) + Q(ii)/2;

    % estimate the rate
    d = hist(X_tilde_pcm, 0:255);
    p = d / length(X_tilde_pcm);
    p = p(p ~= 0);
    R_pcm(ii) = -sum(p .* log2(p));

    % compute PSNR
    MSE = var(X - X_tilde_pcm);
    PSNR_pcm(ii) = pow2db(var(X_tilde_pcm)/MSE);
end

%% c) DPCM encode the pixel values
R_dpcm = zeros(length(Q),1);
PSNR_dpcm = zeros(length(Q),1);
for ii = 1:length(Q)
    X_tilde_dpcm = zeros(size(X));
    d_tilde = zeros(size(X));

    % first sample
    X_tilde_dpcm(1) = Q(ii) * floor(X(1)/Q(ii)) + Q(ii)/2;
    d_tilde(1) = 0;

    % next sample: predictor = past sample
    for nn = 2:length(X)
        d = X(nn) - X_tilde_dpcm(nn-1);
        d_tilde(nn) = Q(ii) * floor(d/Q(ii));
        X_tilde_dpcm(nn) = d_tilde(nn) + X_tilde_dpcm(nn-1);
    end
    % figure
    % imshow(reshape(uint8(X_tilde_dpcm), nR, nC)), title('DPCM encoded image')

    % estimate the rate
    d = hist(d_tilde, min(d_tilde):max(d_tilde));
    p = d / length(d_tilde);
    p = p(p ~= 0);
    R_dpcm(ii) = -sum(p .* log2(p));

    % compute PSNR
    MSE = var(X - X_tilde_dpcm);
    PSNR_dpcm(ii) = pow2db(var(X_tilde_dpcm)/MSE);
end

%% d) Plot the rate-distortion function in both cases. Use the entropy of the quantization symbols as an estimate for the rate.
figure
plot(R_pcm, PSNR_pcm, 'ro-'), hold on
plot(R_dpcm, PSNR_dpcm, 'bo-'), hold off
xlabel('Rate'), ylabel('PSNR [dB]')
legend('PCM', 'DPCM', 'Location', 'BestOutside')
```